

Entity–Memory Graph Retrieval Improves Evidence Coverage in Long-Conversation Question Answering

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Abstract

Entity–Memory graph retrieval keeps dialogue turns as verbatim Memory nodes, links repeated mentions through shared Entities, and connects adjacent Memories with directed chronological edges. At query time the retriever moves from Entity gating through semantic fusion and one-hop chronological recovery to dense backfill. The path can keep a neighboring Memory that dense cosine ranking would otherwise omit. A matched dense control shares the Memory and query vectors, context budget, requested answer protocol, and evaluator, isolating graph structure from changes to the reader.

On 1,986 questions from ten LoCoMo conversations, graph retrieval raises official evidence recall at *top-k* 25 from 79.7468% to 84.4842%. The recall advantage is supported from *top-k* 5 to 50, while no matched cutoff supports an overall final-answer F1 difference. Four paper-eligible requested configurations support empirical robustness across the tested GPT-3.5 and DeepSeek extractors on both outcomes. Embedding robustness is mixed: F1 has no supported contrast, but recall is sensitive to the embedding artifact. The comparison isolates a retrieval-coverage gain from graph structure. It does not establish a final-answer F1 gain, model or embedding equivalence, or cross-dataset generalization.

Keywords: long-term conversational memory; Entity–Memory graph retrieval; evidence retrieval; LoCoMo

1 Introduction

Conversational agents accumulate facts, events, preferences, and relationships over interactions that exceed a model’s usable context. Long-term memory systems must therefore decide what to store, how to organize it, and which parts to retrieve for a new question. The LoCoMo benchmark makes this problem concrete through long, multi-session conversations and questions requiring single-hop, multi-hop, temporal, commonsense, and adversarial reasoning [Maharana et al., 2024]. Long-context and retrieval-augmented generation improve memory-related question answering on LoCoMo, but long-range temporal and causal reasoning remain difficult.

Dense retrieval provides a natural control. It encodes each conversational Memory item, scores that item against the question, and sends the highest-ranked items to an answer model. The design is simple and scalable, but it treats Memory items as independent. A graph can represent repeated entities and

chronological adjacency, allowing retrieval to use structure that is absent from a flat index. A valid comparison must distinguish that structural contribution from changes in embeddings, context budgets, answer models, or evaluation.

We evaluate an Entity–Memory graph built only from conversation content. Each dialogue item becomes a Memory node, normalized Entities become shared nodes, and chronological links connect neighboring Memories. Entity matching is fused with semantic Memory similarity and chronological expansion at query time. Test questions form retrieval queries only. Questions, answers, evidence labels, categories, predictions, and evaluator output are excluded from graph construction.

The study addresses three research questions:

- RQ1.** When Memory nodes, ordered query vectors, answer generation, and evaluation are fixed, does complete graph retrieval improve evidence recall over matched dense retrieval, and does any retrieval gain translate into higher overall final-answer F1?
- RQ2.** Which retrieval components are associated with the observed difference in evidence recall?
- RQ3.** How robust is the graph-retrieval advantage across context cutoffs and input profiles, how sensitive are F1 and evidence recall to fusion and sequence settings and paper-eligible requested extractor/embedding configurations, and where are the category-level strengths and weaknesses concentrated?

These questions motivate three contributions:

- (1) a formal conversation-only Entity–Memory retrieval model that separates graph construction from question-time retrieval;
- (2) an all-question matched comparison that holds Memory vectors, ordered query vectors, answer generation, and evaluation fixed while measuring evidence recall separately from final-answer F1; and
- (3) outcome-specific analyses of components, cutoffs, input profiles, parameters, the four paper-eligible requested configurations, descriptive category profiles, audited cases, and cold/warm operation that delimit where the retrieval difference holds and whether it reaches final-answer F1.

Together, these contributions isolate the retrieval effect of graph structure from changes to the reader or evaluator. The full experiment covers the primary matched A/B comparison, structural ablations, four context cutoffs, raw and time-annotated inputs, fusion/sequence sensitivity, and a paper-eligible two-requested-extractor-by-two-embedding matrix. The main pat-

tern is outcome-specific: graph structure improves evidence coverage, the tested requested extractors support empirical robustness in both F1 and recall, and the tested embeddings yield mixed robustness, with empirical F1 robustness but supported recall differences.

2 Related Work

2.1 Why long conversations need structured retrieval

Long conversations scatter related facts across sessions, so a memory system must preserve both semantic relevance and the relations among separated turns. Retrieval-augmented generation combines a parametric generator with retrieved non-parametric evidence [Lewis et al., 2020]. Dense Passage Retrieval showed that a learned dual encoder can retrieve candidate passages for open-domain question answering using dense representations [Karpukhin et al., 2020]. These studies establish the flat dense-retrieval control used in our matched comparison.

Later systems organize long inputs above the individual-chunk level. RAPTOR recursively embeds, clusters, and summarizes text into a retrieval tree [Sarathi et al., 2024]. MemoRAG uses a long-range model to form global memory and generate clues for a downstream retriever [Qian et al., 2024]. Our method also introduces structure above independent Memory vectors, but it retains dialogue items as the answer evidence rather than replacing them with generated summaries.

2.2 Long-term conversational memory

LoCoMo introduced a human-verified benchmark of long, persona-grounded, multimodal conversations and evaluates question answering, event summarization, and dialogue generation [Maharana et al., 2024]. LongMemEval later framed assistant memory as indexing, retrieval, and reading, and evaluates information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention [Wu et al., 2025]. These benchmarks motivate pipeline-level evaluation rather than attributing all downstream behavior to the language model.

Early agent-memory architectures store experience for later retrieval and reflection. Generative Agents records observations, derives higher-level reflections, and retrieves memories for planning [Park et al., 2023]. MemoryBank adds persistent conversational memory with retrieval, updating, and time-dependent forgetting [Zhong et al., 2024]. MemGPT instead manages limited context through virtual memory and hierarchical tiers [Packer et al., 2023].

Recent systems place more structure in the memory store. Mem0 extracts, consolidates, and retrieves salient conversational information, with a graph variant for relational memory [Chhikara et al., 2025]. A-Mem dynamically indexes and links structured memory notes into an evolving network [Xu et al., 2025]. Our study examines a narrower question than these end-to-end memory architectures. Given a fixed requested reader protocol and fixed Memory representations, we test whether a

conversation-derived graph changes evidence retrieval relative to a flat dense control.

2.3 Graph retrieval and the comparison gap

Graph retrieval can expose relations that independent text chunks omit. HippoRAG combines a knowledge graph with Personalized PageRank for long-term knowledge integration and multi-hop retrieval [Gutiérrez et al., 2024]. GraphRAG constructs a graph index and community summaries for global questions over document collections [Edge et al., 2024]. GRAG retrieves textual subgraphs and supplies textual and topological views to a generator [Hu et al., 2025], while LightRAG combines graph indexing with dual-level retrieval over entities and relations [Guo et al., 2025].

Graph structure has also become central to conversational memory. APEX-MEM stores temporally grounded events in a property graph and resolves evolving information at retrieval time [Banerjee et al., 2026]. Mnemis combines a base similarity graph with top-down selection over a semantic hierarchy [Tang et al., 2026]. GAM separates an event-progression graph from a topic-association network [Wu et al., 2026], whereas TiMem consolidates conversation observations into a temporal hierarchy [Li et al., 2026]. Our graph is smaller in scope. Shared Entity nodes link mentions across dialogue-item Memories, and chronological edges preserve local sequence. These systems motivate graph memory, but they also expose an attribution problem: system-level comparisons often change the graph, encoder, reader, prompt, and context budget together. DRAGON, for example, is a dense retriever trained with diverse data augmentation [Lin et al., 2023] and is used in the original LoCoMo RAG setup. We therefore treat external systems as descriptive anchors and use an internal matched dense control to isolate graph retrieval. This is an experimental-control choice, not a new evaluation metric.

3 Method

3.1 Task and estimands

Let a conversation be an ordered collection of Memory items $\mathcal{M} = \{m_1, \dots, m_n\}$, where each item contains a dialogue identifier, speaker, text, timestamp, and optional image caption. Given a question q , a retriever returns an ordered context $R_k(q) \subseteq \mathcal{M}$ with $|R_k(q)| = k$. The study tests whether a conversation-derived graph changes the evidence contained in $R_k(q)$ when the Memory representations, query vectors, reader, and evaluator are fixed.

Graph construction is a function of conversation content only:

$$G = B(\mathcal{M}) = (V_M \cup V_E, E_{EM} \cup E_{MM}). \quad (1)$$

Here, V_M contains one node per Memory item, V_E contains normalized Entities shared across items, E_{EM} contains Entity–Memory mention links, and E_{MM} contains directed next/previous links between chronologically adjacent Memories. The constructor B has no access to questions, answers,

evidence annotations, category labels, predictions, or evaluator output. Questions enter only through the retrieval function $R_k(q; G)$.

The primary estimand is the paired difference in official evidence recall between graph retrieval and matched dense retrieval. Final-answer F1 is a separate downstream outcome. This distinction prevents additional retrieved evidence from being interpreted automatically as an improvement in generated answers.

3.2 Conversation-only graph construction

Figure 1(a) is the matched dense RAG control: independent Memory ranking with no Entity or chronological signal. Figure 1(b) shows the conversation-only graph and the progressive retrieval path. For each conversation, the constructor consumes session timestamps, dialogue identifiers, speaker names, dialogue text, and image captions. Each normalized dialogue item becomes a Memory node. A gpt-3.5-turbo extractor at temperature 0.3 identifies Entities from the conversation text. Entity strings are normalized, deduplicated case-insensitively, and connected to every Memory in which they occur. Deterministic speaker links add the dialogue speaker as an Entity. Adjacent Memory nodes are connected in chronological order.

The complete ten-conversation graph contains 5,882 Memory nodes. Graph construction does not consume QA questions, gold answers, evidence annotations, category labels, judge outputs, previous predictions, or question-driven ledgers. This separation prevents test questions from shaping the stored memory structure.

3.3 Entity-gated retrieval with chronological recovery

The question supplies two retrieval-only signals. A question-Entity extractor produces a set of normalized keys $Q_E(q)$, and the question vector is loaded from a complete artifact generated with OpenAI’s text-embedding-3-small model [OpenAI, n.d.]. The stored artifact is immutable and bound to the dataset hash, ordered question digests, model, role, vector dimension, normalization, and artifact SHA-256. Formal retrieval fails on a cache miss and makes no live query-embedding request.

Let $\rho(e, u) \in [0, 1]$ be the peak-normalized BM25 score between Entity node e and question key u . Exact normalized-key equality receives a score of one. Matches below threshold $\tau = 0.5$ are removed, and at most 20 Entity nodes are retained for each key. For the set $Q_E^+(q)$ of keys with at least one retained match, the degree-discounted Entity score is

$$h(e, q) = \frac{1}{|Q_E^+(q)|} \sum_{u \in Q_E^+(q)} \rho(e, u) \times \frac{1}{\log(1 + \deg(e))}. \quad (2)$$

For Memory m , let $c(m, q)$ and $w(m, q)$ be the maximum incident scores from content Entities and person/speaker Entities, respectively. The Memory seed score uses $c_m = c(m, q)$,

$w_m = w(m, q)$, and $\delta = 0.25$:

$$s_e^{(0)}(m, q) = \begin{cases} \max\{c_m, w_m\}, & c_m > 0, \\ \delta w_m, & c_m = 0. \end{cases} \quad (3)$$

Thus, a speaker match is dampened only when no content Entity supports the same Memory.

One-hop chronological expansion transfers each seed score to its immediately preceding and following Memories. With $N_{\text{seq}}(m)$ denoting those neighbors, the expanded Entity score is

$$s_e(m, q) = \max \left\{ s_e^{(0)}(m, q), \lambda \max_{m' \in N_{\text{seq}}(m)} s_e^{(0)}(m', q) \right\}. \quad (4)$$

In the experiments, we generally set λ to 0.5; the preregistered sensitivity analysis also tests 0.25 and 1.0.

The nonempty Entity scores define the gated candidate set $C(q) = \{m : s_e(m, q) > 0\}$. If this set is empty, the candidate set is all Memories. Within the active set, the primary condition combines the Entity score with signed cosine similarity $s_s(m, q)$:

$$s(m, q) = \alpha s_e(m, q) + \beta s_s(m, q). \quad (5)$$

The primary setting uses $\alpha = 0.30$ and $\beta = 0.70$. Candidates are ordered by decreasing $s(m, q)$ and then by dialogue id. When the gated set contains fewer than k Memories, unused items from full-pool dense retrieval fill the remaining positions without displacing gated items.

The primary context budget (*top-k*) is 25. Cutoff analyses use *top-k* 5, 10, and 50. The dense control ranks the same Memory representations with the same semantic query vectors but does not use Entity matching or sequence expansion. A dense graph-construction control, B_{embed} , verifies that graph construction alone does not change dense retrieval.

4 Experimental Setup

4.1 Matched conditions and comparison boundary

The primary comparison holds constant the dataset and row order, Memory text, Memory vectors, immutable ordered query vectors, answer prompt, answer model, token budget, evaluation package, output isolation, and aggregation. The treatment activates Entity matching, graph fusion, and chronological expansion. The B_{embed} control must reproduce the dense control’s ordered context identifiers for all rows before the primary graph contrast is interpreted.

The original LoCoMo RAG setup uses DRAGON, but our pinned upstream reproduction did not pass the declared tolerance. The local raw-DRAGON condition is therefore diagnostic and is excluded from the matched and inferential comparisons. The primary evidence concerns A, B_{embed} , and B under the shared artifacts defined above.

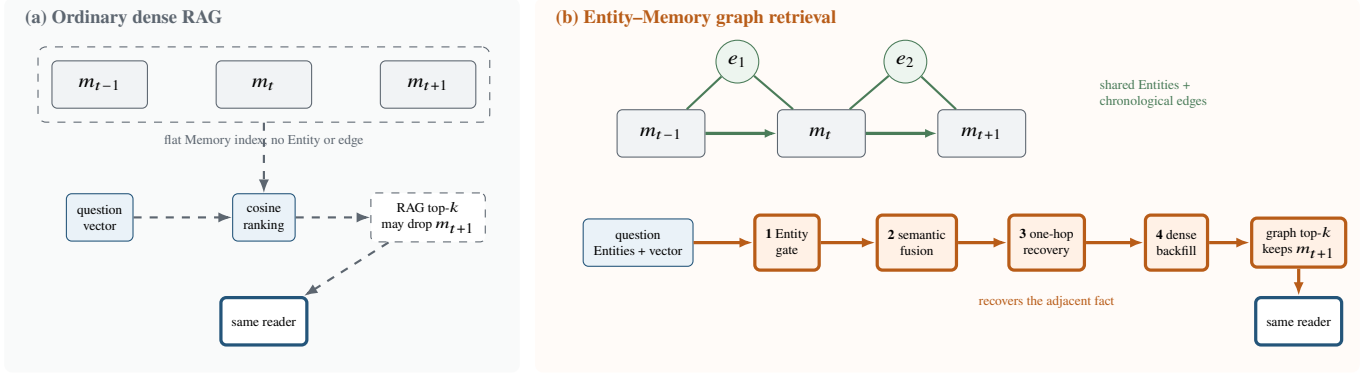


Figure 1: Matched contrast between ordinary dense RAG and Entity-Memory graph retrieval. (a) The control ranks independent Memory vectors by cosine similarity and sends the top- k items to the reader; it has no Entity nodes and no chronological edges. (b) Conversation turns first become a graph with shared Entities and directed Memory edges; retrieval then gates, fuses, recovers one-hop neighbors, and backfills. Both paths share Memory vectors, the query vector, $top-k$, the requested answer protocol, and the frozen evaluator. The neighbor annotation is schematic and does not report a metric. The control is the internal matched dense condition, not the official LoCoMo DRAGON setup.

Each metric-bearing condition writes to an absent, condition-specific output directory. Its record includes the source commit, resolved configuration, model names, artifact identities, query-cache hits and misses, prompt budgets, and graph compliance. The vendored evaluator is verified against its SHA-256 manifest before formal use.

4.2 Ablation study design

The design matrix separates structural attribution, operating-condition stress tests, local parameter sensitivity, and model substitution. Within each family, the listed treatment changes while the comparison boundary retains the shared artifacts and evaluation protocol; the final row records a completed condition that is useful diagnostically but not eligible for model-only inference.

4.3 Evaluation protocol and statistical design

Retrieved Memory text and identifiers enter the frozen LoCoMo-aligned answer interface. The requested answer model is `gpt-3.5-turbo`, with a system-role prompt, temperature 0, one question per batch, and a 32-token completion limit. The immutable evaluation package retains its category-specific generation, decoding, F1, evidence-recall, rounding, and aggregation behavior. Overall F1 measures final-answer overlap, whereas `recall_acc` is the official evidence-recall definition. The repository-defined Categories 1–4 subset F1 is reported only as a diagnostic, not as an official LoCoMo metric.

Every formal condition contains the same 1,986 QA rows from ten conversations. Matched differences are estimated with 10,000 paired-question bootstrap resamples and 10,000 resamples of whole conversations, using seed 20260727. The conversation-cluster estimator reflects uncertainty over the ten conversation units. Component, input, fusion, and sequence families use Holm step-down correction within each outcome and estimator. A difference is treated as supported only when the preregistered evidence gate passes. Failure to reject is not interpreted as equivalence.

The paper-eligible configuration matrix crosses two requested Entity extractors with two embedding aliases and uses 10,000 whole-conversation bootstrap resamples with seed 20260814. Holm correction is applied separately to extraction/F1, extraction/recall, embedding/F1, and embedding/recall. The preregistered sequential stop gate uses an absolute three-point band and is triggered only when a 95% cluster interval lies wholly below -3 points or wholly above $+3$ points. We report empirical robustness only as an outcome-specific interpretation when a tested substitution has no Holm-corrected rejection and does not trigger this gate; this is not a formal equivalence test. Two completed `gpt-5-mini` cells are retained only as diagnostic observations: their frozen records specify a 2,500-token extraction limit, whereas the executed model-specific path used 4,000 tokens and 12 calls exceeded 2,500. They are therefore excluded from the model-only robustness claims and category ranges. For all F1 contrasts, the requested GPT-3.5 alias, prompt, role, temperature, and token budget are fixed, but the provider-returned deployment revision was not recorded and Category-5 option order remains unseeded. We therefore treat F1 as an observed downstream outcome under a matched requested protocol; `recall_acc` provides the cleaner retrieval-stage comparison.

5 Results

5.1 Primary matched comparison

Complete graph retrieval improves `recall_acc` over the matched dense control by 4.7374 percentage points (Table 1). The paired-question 95% interval is 3.6504–5.8425 points, and the conversation-cluster interval is 3.5286–6.0124 points. Both two-sided p -values are 0.0002. Overall F1 changes by 0.4998 points, with paired and cluster intervals of -0.5745 – 1.5514 and -0.1773 – 1.2711 points. The F1 difference is not supported by either estimator.

The dense graph-construction control and A have zero ordered-context mismatches across all 1,986 rows and identical

Ablation and robustness design. Each row changes only the listed factor within its declared comparison; all conditions retain the same requested answer protocol and frozen evaluator.

Family	Treatment and control	Fixed factors	Purpose
Structural components	Remove sequence expansion, Entity-score fusion, semantic scoring, or speaker links from complete B	Dataset, Memory text, primary vectors, <i>top-k</i> 25, requested reader protocol, frozen evaluator	Attribute the primary retrieval difference
Cutoff and input	Matched A/B at <i>top-k</i> 5, 10, 25, 50; raw-text versus time-annotated input	Paired rows and within-profile artifacts	Test context budget and input-profile dependence
Parameters	Entity/Semantic weights 0.10/0.90, 0.30/0.70, 0.50/0.50; sequence scale 0.25, 0.5, 1.0	Complete-B graph and reader	Bound local sensitivity without claiming universal optima
Robustness across configurations	Requested gpt-3.5-turbo or deepseek-v4-flash Entity extraction crossed with TES or Doubao embeddings	Complete B, <i>top-k</i> 25, requested extraction budget 2,500, requested GPT-3.5 reader protocol	Test extractor and embedding substitutions in a paper-eligible 2×2 matrix with outcome-specific robustness claims
Diagnostic only	Two gpt-5-mini extraction cells	Reader and retrieval settings fixed, but runtime extraction budget was 4,000 rather than the frozen 2,500	Excluded from model-only inference pending rerun

Table 1: Primary matched comparison at *top-k* 25. Values are percentages.

Condition	Overall F1	recall_acc
A (dense Memory)	42.0681	79.7468
B_{embed} (dense graph control)	42.0677	79.7468
B (complete graph retrieval)	42.5680	84.4842

evidence recall. Their F1 values differ by less than 0.001 percentage points, consistent with frozen Category-5 option-order randomness rather than a retrieval difference.

5.2 Component analysis

The component analysis associates several graph operations with the retrieval gain (Table 2). Complete B improves recall over the no-sequence condition by 1.6728 points and over the no-Entity-score-fusion condition by 4.7626 points after Holm correction. Neither contrast supports F1. Sequence expansion after Entity gating contributes 0.4364 recall points. The semantic channel relative to Entity-only retrieval contributes 17.4634 recall points and 5.0780 F1 points, with both outcomes supported within the component family. Removing deterministic speaker links lowers recall by 1.4822 points without a supported F1 change.

The apparently opposite signs for “sequence expansion after Entity gating” reflect different outcomes, not a contradiction. For the Category-4 question “How long have Mel and her husband been married?”, the gated control misses gold turn D3:16 and produces an incorrect verbose sentence that nevertheless shares enough tokens with the reference to receive

F1 0.429. Sequence expansion retrieves D3:16 and yields the correct concise answer “5 years”, raising evidence recall from 0 to 1 but receiving F1 0.364. This case illustrates why the aggregate +0.4364 recall-point contrast can coexist with an unsupported −0.0609 F1-point contrast: token overlap and evidence coverage measure different stages of the pipeline.

5.3 Cutoff and input-profile robustness

The evidence-recall advantage persists across all preregistered cutoffs (Table 3). B minus A recall differences range from 3.6159 to 5.5702 points. Paired-question and conversation-cluster intervals are above zero at every cutoff. Both matched F1 intervals include zero at every cutoff. The cutoff analysis therefore supports an evidence-coverage advantage over the tested range, not an answer-quality advantage.

The input-profile contrasts preserve the distinction between evidence recall and final-answer F1 (Table 2). Time annotation has no supported within-B overall effect after correction. When both A and B use raw text, B retains a supported 4.9522-point recall advantage without an F1 advantage.

5.4 Parameter sensitivity

The parameter results bound the component findings in Table 2. Reducing Entity/Semantic weights from 0.30/0.70 to 0.10/0.90 lowers recall by 2.3062 points after correction. Equal 0.50/0.50 weights are not distinguishable from the primary setting. No overall F1 or recall contrast among sequence scales 0.25, 0.5, and 1.0 survives family correction.

Table 2: Component and input contrasts at *top-k* 25. Differences are treatment – control in percentage points. “Supported” refers to the preregistered family-corrected inference unless noted otherwise.

Contrast	$\Delta F1$	$\Delta \text{recall_acc}$	Conclusion
B – B without sequence expansion	+0.5137	+1.6728	Recall supported; F1 not supported
B – B without Entity-score fusion	+0.8014	+4.7626	Recall supported; F1 not supported
Sequence expansion after Entity gating	−0.0609	+0.4364	Recall supported; F1 not supported
B – Entity-only retrieval	+5.0780	+17.4634	Both outcomes supported
Raw-text B – annotated-input B	+0.2407	−0.2472	Neither outcome supported
Raw-text B – raw-text A	+0.3508	+4.9522	Recall supported; F1 not supported
B – B without speaker links	+0.2415	+1.4822	Recall supported; F1 not supported

Table 3: Matched cutoff analysis. Values are percentages; differences are B minus A in percentage points. Every paired-question and conversation-cluster recall interval is above zero. Every corresponding F1 interval includes zero.

<i>top-k</i>	A F1	B F1	$\Delta F1$	A recall	B recall	Δrecall	Paired recall CI	Cluster recall CI
5	39.9053	39.8264	−0.0790	59.3584	64.1667	+4.8083	[3.3450, 6.2808]	[2.8809, 6.6134]
10	41.7607	42.1412	+0.3805	68.9145	74.4847	+5.5702	[4.2928, 6.8624]	[4.2365, 6.7522]
25	42.0681	42.5680	+0.4998	79.7468	84.4842	+4.7374	[3.6504, 5.8425]	[3.5286, 6.0124]
50	42.0782	41.7832	−0.2950	86.7148	90.3306	+3.6159	[2.7754, 4.4735]	[2.4603, 4.6426]

Table 4: Paper-eligible 2×2 configuration audit for complete B at *top-k* 25. TES is text-embedding-3-small; DB is doubao-embedding-vision. Values are percentages.

Requested extractor	Embedding alias	F1	recall_acc
GPT-3.5	TES	42.5680	84.4842
DeepSeek v4 Flash	TES	42.1087	84.6005
GPT-3.5	DB	42.1507	82.1978
DeepSeek v4 Flash	DB	42.8158	82.5778

5.5 Outcome-specific robustness across requested configurations and descriptive category profile

Across the four protocol-matched cells in Table 4, F1 spans 42.1087–42.8158. Neither the two requested-extractor F1 contrasts nor their recall contrasts reject zero after within-family Holm correction. These results support limited empirical robustness of both outcomes across the tested GPT-3.5 and DeepSeek requested extraction configurations, without establishing strict extractor equivalence. The corresponding embedding F1 contrasts also do not reject zero, but the DB frozen artifacts lower recall by 2.0227 points under DeepSeek extraction and 2.2864 points under GPT-3.5 extraction. Both recall contrasts remain supported after correction. Each 95% cluster interval contains values on both sides of the preregistered −3-point boundary, so neither interval triggers the gate for material robustness failure. Passing that gate does not establish embedding equivalence. Robustness across the tested embeddings is therefore outcome-specific: F1 supports empirical robustness under the declared rule, whereas evidence recall is sensitive to the frozen embedding artifacts.

Row-level examples show why the aggregate conclusion must remain outcome-specific. With TES fixed, replacing GPT-3.5 extraction by DeepSeek on “What gifts has Deborah received?” recovers one of five annotated turns (recall 0 to 0.2), yet the independently generated answer changes from “A bouquet from a friend” to “A bouquet” and token F1 falls from 0.44 to 0, partly because the gold answer misspells *bouquet*. Conversely, with the DeepSeek graph fixed, TES retrieves both annotated turns for the Voyageurs National Park question (recall 1), whereas DB drops one (recall 0.5). DB’s incorrect “Unnamed national park” nevertheless receives F1 0.667, above the TES answer’s 0.138, because it shares a larger fraction of tokens with the gold name. These cases do not establish single-row causal effects; they document how stable aggregate F1 can coexist with observable retrieval changes.

Concrete rows illustrate this profile. On the temporal question “Which classes did Evan join in mid-August 2023?”, dense A misses D8:12 and answers “winter activities” (recall 0, F1 0.2). B retrieves D8:12 together with its D8:13–14 neighbors and answers “Painting classes” (recall 1, F1 1.0). For the single-hop question asking what keeps Evan busy while his knee heals, A retrieves nearby D11 turns but omits D11:6 and answers “Swimming”; B adds D11:6 at rank 19 and answers “Watercolor painting”, moving both outcomes from 0 to 1. These cases are consistent with Entity links locating the relevant session and chronological recovery adding the exact adjacent fact; they do not isolate either operation as the cause of a single-row outcome.

The weaker categories exhibit different retrieval and answer patterns. For the open-domain Dr. Seuss question, both A and B retrieve the annotated evidence that Caroline collects classic children’s books, but their answers omit the reference rationale and receive F1 0.222 and 0.118. In these stored

Table 5: Category profile across the four paper-eligible configurations, with descriptive external anchors from the original LoCoMo Dialog@25 condition [Maharana et al., 2024] and LightGMEM [Anonymous, 2026]. External readers, embeddings, and category scope differ, so the table does not establish cross-paper superiority.

Category	B F1 range	Dialog F1	LightGMEM F1	B recall range	Dialog recall	Descriptive position
Multi-hop	37.6–39.9	38.7	41.9	61.5–67.3	62.5	B F1 range overlaps the Dialog anchor
Temporal	40.2–43.1	37.2	56.4	89.2–90.2	83.5	B recall is numerically above the Dialog anchor
Open-domain	17.3–18.6	25.0	27.2	48.4–56.4	52.6	Lowest B answer F1 range among Categories 1–4
Single-hop	63.8–64.1	59.9	64.1	91.1–92.7	87.1	Highest B answer F1 range
Adversarial	8.3–11.2	12.8	—	80.2–82.6	66.3	B recall is numerically above Dialog; raw B F1 is lowest under the separate Category-5 branch

outputs, retrieving the annotated evidence is insufficient to match the reference rationale. Conversely, for the adversarial question about what inspired Melanie’s art-show painting, B retrieves D9:16 whereas A does not, yet both stored outputs receive F1 0 under the frozen Category-5 branch. This row is consistent with the aggregate pattern of higher adversarial recall without an answer-quality advantage; it does not identify the generation or scoring branch as the cause.

5.6 Operational cost under cold and warm retrieval

The warm replay makes no new provider requests in the three cache-sensitive stages (Table 6). The cold-to-warm retrieval speedup is 309.48×. Cold and warm each cover the same 44 ordered batches, with 1,998 reads from the immutable query artifact, zero misses, and zero live query embeddings. This includes a 1.29-second warm Memory-index cache load. The zero-request result applies to replaying the same question set with complete graph, index, query-vector, and question-Entity caches; unseen questions may require new query embedding and question-Entity calls. The formal answer trace contains 1,986 requests, 2,744,099 input tokens, and 16,269 output tokens. Because answer generation is independent of cache warmth, the same validated trace is attached to each state rather than rerun.

Graphs occupy 21,689,931 bytes, Memory indexes 29,946,660 bytes, and Entity/question caches 3,411,176 bytes. Including the attached answer trace, the measured cold path uses 7,903,980 input and 546,178 output tokens, or 8,450,158 total tokens. Amortized over 1,986 questions, this is 4,254.9 tokens per question; the answer stage accounts for 1,389.9 tokens per question. Applying OpenAI public list prices accessed on 1 August 2026 gives an illustrative total of \$4.67, or \$0.00235 per question.¹ This conversion excludes the separately built read-only query-vector artifact, storage and platform fees, and

unrecorded external charges. It is not a frozen experimental metric.

6 Discussion

The experiments separate evidence coverage from answer quality. Entity links, semantic scoring, and chronological expansion retrieve more annotated evidence than the matched dense control, and the difference persists across four context budgets. Under the matched requested answer protocol, the corresponding overall F1 difference is not supported. Evaluations that report only answer scores can therefore obscure retrieval changes, while retrieval improvements should not be described as end-to-end quality gains.

This separation complements LongMemEval’s decomposition of assistant memory into indexing, retrieval, and reading stages [Wu et al., 2025]. Systems such as Mem0, A-Mem, APEX-MEM, Mnemis, GAM, and TiMem jointly change several of these stages [Chhikara et al., 2025, Xu et al., 2025, Banerjee et al., 2026, Tang et al., 2026, Wu et al., 2026, Li et al., 2026]. Their published scores therefore answer broader system-level questions than our matched contrast. Differences in readers, graph construction, prompts, context budgets, and metrics preclude a numerical ranking against our result. Our evidence instead identifies what changes when the stored Memory text, Memory vectors, query vectors, requested answer protocol, and evaluator remain fixed.

The component results suggest that the complete difference is distributed across the retrieval path. The semantic channel is the largest isolated contributor. Entity fusion, sequence expansion, and speaker links add smaller supported recall contributions. The raw-text comparison indicates that the B-over-A recall advantage does not depend on the tested time annotation. Sensitivity results also discourage presenting the primary parameters as universally optimal. One lower-Entity-weight condition reduces recall, an equal-weight condition is not distinguishable, and no tested sequence scale survives family correction.

The configuration audit provides positive but bounded robustness evidence. Across the four paper-eligible requested con-

¹Prices used for this contextual conversion were \$0.50/\$1.50 per million GPT-3.5 Turbo input/output tokens and \$0.02 per million text-embedding-3-small tokens: <https://developers.openai.com/api/docs/models/gpt-3.5-turbo> and <https://developers.openai.com/api/docs/models/text-embedding-3-small>.

Table 6: Primary-B cold and warm measurement, separated into provider work and retrieval latency. Cold stage times overlap and must not be summed as end-to-end latency.

Provider stage	Cold req.	Input tokens	Output tokens	Cold call time	Warm req.
Conversation Entity extraction	5,873	3,731,393	425,637	9,792.76 s	0
Memory embedding	591	214,229	0	443.38 s	0
Question Entity extraction	1,974	1,214,259	104,272	2,970.83 s	0

Retrieval state (1,986 QA)	Provider req.	Mean per QA	p95 per QA
Cold	0	1.5123 s	2.6132 s
Warm	0	0.004886 s	0.006270 s

figurations, no requested-extractor F1 or recall contrast rejects zero after Holm correction. This supports limited empirical robustness across the tested GPT-3.5 and DeepSeek requested extraction configurations, but it does not establish extractor equivalence, independence from extraction, or arbitrary-model invariance. The embedding result is mixed: F1 has no supported contrast, whereas recall falls for the Doubao frozen artifacts relative to TES under the fixed, uncalibrated 0.30/0.70 fusion. Robustness therefore applies to a specified outcome and tested configuration, not to unrestricted model exchange without retuning or validation.

The category profile and cases are consistent with, but do not causally identify, the intended retrieval mechanisms. In the audited single-hop and temporal rows, Entity matching locates a relevant session and one-hop edges add an adjacent turn. Open-domain rows can contain the annotated evidence while their generated answers omit the reference rationale. Category 5 uses a separate frozen option-generation and scoring branch, and the audited row exhibits recall/F1 divergence under that branch. These observations place the highest answer F1 in single-hop questions and higher recall than the original Dialog@25 anchor in temporal and adversarial questions. They do not establish a category-level causal effect or a protocol-matched cross-system advantage.

The cold/warm result exposes an operational tradeoff. Conversation Entity extraction and Memory embedding are one-time costs, while uncached question-Entity extraction dominates cold retrieval latency. Replaying the same question set with complete caches requires no new provider request before answer generation and reduces retrieval to milliseconds per question. This measurement does not imply that unseen questions are free, or that another machine, provider, or workload has the same latency.

Limitations and future work. First, formal evidence comes from ten conversations in one LoCoMo release. Conversation-cluster bootstrap reflects uncertainty over these ten units but does not establish cross-dataset generalization. Replication on additional conversation-memory benchmarks is needed.

Second, no matched cutoff supports an overall final-answer F1 difference. The contribution concerns retrieval coverage, not a general improvement in generated answers. Alternative readers and evidence selectors require separately controlled

experiments.

Third, the official DRAGON reproduction gate did not pass. The local raw DRAGON experiment changes semantic and query artifacts and remains diagnostic. A future official-reference comparison requires parity with the pinned upstream implementation.

Fourth, the requested answer-model name is recorded, but the provider did not persist a separate actual-model identity in the formal snapshots. Category 5 also retains the frozen upstream unseeded option-order behavior. Future runs should preserve provider-returned model identities when available and quantify Category-5 generation variance separately. The robustness artifacts likewise do not fully identify every provider deployment revision for extraction and embedding. The matrix therefore supports outcome-specific empirical robustness across the paper-eligible requested configurations represented by their frozen artifacts, not robustness to fully identified provider deployments; a strict deployment-level claim requires telemetry-complete reruns.

Fifth, the two GPT-5 mini robustness cells have a frozen-versus-runtime extraction-budget mismatch (2,500 versus 4,000 tokens), and 12 calls exceeded the frozen limit. Their observed metrics are diagnostic only. A corrected three-extractor robustness claim requires rebuilding and rerunning those cells with one explicitly resolved budget; future graph-cache identities should also bind extraction temperature and token budget rather than relying on source and artifact hashes for retrospective audit.

Finally, system timing was measured on one machine and one provider endpoint. The warm condition assumes complete reusable graph, index, query-vector, and question-Entity artifacts for the same question set. Token usage is directly measured, whereas the dollar conversion uses public prices accessed after the experiment and is not a frozen metric. Prices, platform fees, and hardware costs can change. Replication across hardware and providers is needed before deployment-level latency claims.

7 Conclusion

Long-conversation question answering requires retrieval that preserves relevant evidence without using test questions to

build memory. Graph retrieval may represent links that a flat index omits, but a fair test must hold the dense representations, query vectors, answer generation, and evaluation fixed. Our matched design addresses this evidence gap.

The matched comparison first finds that the Entity–Memory graph improves official evidence recall over dense Memory retrieval on the evaluated LoCoMo set, whereas the corresponding overall final-answer F1 difference is not supported. Component contrasts then associate the recall difference with semantic scoring, Entity fusion, chronological expansion, and speaker links. Finally, the recall advantage persists across *top-k* 5–50 and the raw-text input profile. Its magnitude is sensitive to the 0.10/0.90 Entity/Semantic fusion setting, while no sequence-scale contrast survives family correction. The paper-eligible requested-configuration matrix supports limited empirical robustness of F1 and evidence recall across the tested GPT-3.5 and DeepSeek extraction configurations. Embedding robustness is outcome-specific: neither embedding F1 contrast rejects zero, whereas Doubao frozen artifacts reduce recall by 2.02–2.29 points relative to TES. The corresponding intervals cross the preregistered –3-point boundary, so the gate for material robustness failure is not triggered, but embedding equivalence is not established. Descriptive category ranges, supplemented by non-protocol external anchors, place the highest answer F1 in single-hop questions, higher evidence recall than Dialog@25 in temporal and adversarial questions, and the lowest answer F1 among Categories 1–4 in open-domain questions; adversarial raw F1 is lower under its separate Category-5 branch. Cold/warm measurements also show that the graph and retrieval artifacts can be reused for identical-question replay, although unseen questions can require new provider calls.

This work is best viewed as a scoped retrieval study. The ten-conversation sample, two paper-eligible requested extractors, two frozen embedding artifacts, failed official DRAGON reproduction gate, single requested reader alias/protocol, and single-environment latency measurement bound the claim. Separating evidence coverage from answer quality provides a transferable evaluation principle for future long-term conversational memory systems.

Reproducibility and Responsible Use

Reproducibility. Every formal condition records its source commit, resolved configuration, dataset and artifact SHA-256 values, isolated output path, query-cache usage, provider-token telemetry, graph-input audit, and prompt budget. Query vectors are complete and read-only during formal retrieval. The evaluation package is vendored and hash-verified. Statistical reports and the final cost report have independent byte-identical reproductions.

Responsible use. The graph uses conversation data, including speaker names and image captions, which can contain personal information in real deployments. Systems applying this method outside the benchmark should establish consent, retention, deletion, and access-control policies. The present

work evaluates benchmark retrieval and does not study privacy attacks or sensitive-memory deletion.

Data and Code Availability

LoCoMo data are available from the benchmark authors [Maharana et al., 2024]. Code, experiment manifests, validation records, and manuscript evidence snapshots for this study are available at https://github.com/Sun668/em_graph_memory/tree/v1.0.8. The exact manuscript snapshot corresponds to the repository history recorded with the arXiv source bundle.

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A Resolved Primary Configuration

Table 7: Resolved settings for the primary B condition.

Dimension	Resolved value
Dataset	LoCoMo ten-conversation release; 1,986 QA rows
Graph inputs	Session time, dialogue id, speaker, text, image caption
Excluded graph inputs	Questions, answers, evidence, categories, judges, predictions
Memory nodes	5,882
Conversation Entity extractor	gpt-3.5-turbo; temperature 0.3
Memory embedding model	text-embedding-3-small
Answer model request	gpt-3.5-turbo; temperature 0; 32 output tokens
Primary context budget	<i>top-k</i> 25
Entity / semantic weights	0.30 / 0.70
Entity relative-score threshold	0.5
Maximum matches per Entity key	20
Speaker-only dampening	0.25
Degree discount	Enabled
Sequence expansion scale	0.5
Query embeddings	Complete read-only artifact; L2-normalized float32
Bootstrap	10,000 paired QA and 10,000 conversation-cluster resamples
Bootstrap seed	20260727

B Validity and Reproducibility Gates

Table 8: Checks applied before promotion of formal evidence.

Gate	Requirement	Outcome
Graph construction	Conversation-only inputs; no QA-derived construction fields	Pass
Evaluator integrity	All 16 vendored files match the frozen SHA-256 manifest	Pass
Output isolation	One absent or empty directory per resolved formal condition	Pass
Query vectors	Complete ordered coverage; zero miss; zero live query embedding	Pass
Dense control	A and B_{embed} ordered contexts equal for all 1,986 rows	Pass
Prompt budget	Non-data scaffold no longer than 5,000 characters	Pass (2,410)
Statistical reproduction	Independent report build is byte-identical	Pass
Cost reproduction	Independent cost report build is byte-identical	Pass
Cache preservation	Quarantined artifacts retain frozen identities	Pass